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NF270.5 data sets

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To: Xinhong Liu <xliu27@nd.edu>
Cc: Jonathan Ouimet <jouimet@nd.edu>

Mon, Oct 2, 2023 at 1:38 PM

Dear Xinhong,

I have been working on fixing the issues you pointed out with some of the data sets for NF270.5, NF270.6, and NF270.7.

1. NF270.5 There were four experiments where conductivity data was missing and the "lag" or "overflow" mode could not be determined (11.2.22overflow, 11.2.22Rev, 11.3.22RevB, 11.3.22B). I reran these experiments since I was not able to fill in the missing conductivity data from those past experiments (9.25.23overflow, 9.25.23reverse, 9.26.23reverse, 9.26.23local). Attached are the new data sets and updated excel sheets. These experiments were run using a new probe, therefore, I have also attached the information regarding the new calibration curve. This is the calibration curve for KCl we will be using moving forward. The NF270.6 and NF270.7 data below should be used with the old calibration curve though.

2. NF270.6 The global experiment (15mM KCl feed, 120mM KCl diafiltrate) ran on 1.30.23 was missing conductivity signals, and ICP data did not match conductivity probe data. At the time these experiments were run, we created a file that stored all conductivity readings data in its own text file (something that was not available to us for the NF270.5 membranes, which is why I had to rerun them). Using this, I went back and filled in all possible "#NUM!" values that were missing. The updated text file is attached. I also ran the feed, diafiltrate, final tube, and retentate samples again on ICP. I have updated the values and it appears that the retentate concentration now matches the conductivity reading (conductivity = 101.19mM, ICP = 99.94mM). The updated excel files for that are also attached.

3. NF270.7 For this global experiment (15mM KCl feed, 120mM KCl diafiltrate) ran on 1.30.23, similar to the NF270.6, I went ahead and attached an updated text file substituting all possible "#NUM!" values with the conductivity recordings stored in the separate conductivity text file. Additionally, I ran ICP on the feed, diafiltrate, final tube, and retentate again. The conductivity retentate reading (102.98mM) now better aligns with the ICP reading (101.83mM). This updated excel sheet is also attached.

Please let me know if there are still any inconsistencies and we can discuss them on our own time, that way we do not need to take any of Dr. Dowling and Dr. Phillip's time in the Monday meetings.

Best,
Laurianne

[Quoted text hidden]

10 attachments

-  **Exp_01302023_NF270_6_lag_D_SS_KCl.txt**
18K
-  **Exp_01302023_NF270_7_lag_D_SS_KCl.txt**
20K
-  **Membrane.NF270.6.xlsx**
37K
-  **Membrane.NF270.7.xlsx**
36K
-  **Exp_09262023_NF270_5_reverse_D_SS_KCl.txt**
22K
-  **Exp_09252023_NF270_5_reverse_D_SS_KCl.txt**
18K
-  **Exp_09252023_NF270_5_overflow_D_SS_KCl.txt**
22K



Exp_09262023_NF270_5_local_D_SS_KCl.txt
23K



Membrane.NF270.5.xlsx
125K



Xinhong -Diafiltration KCl Calibration.xlsx
18K